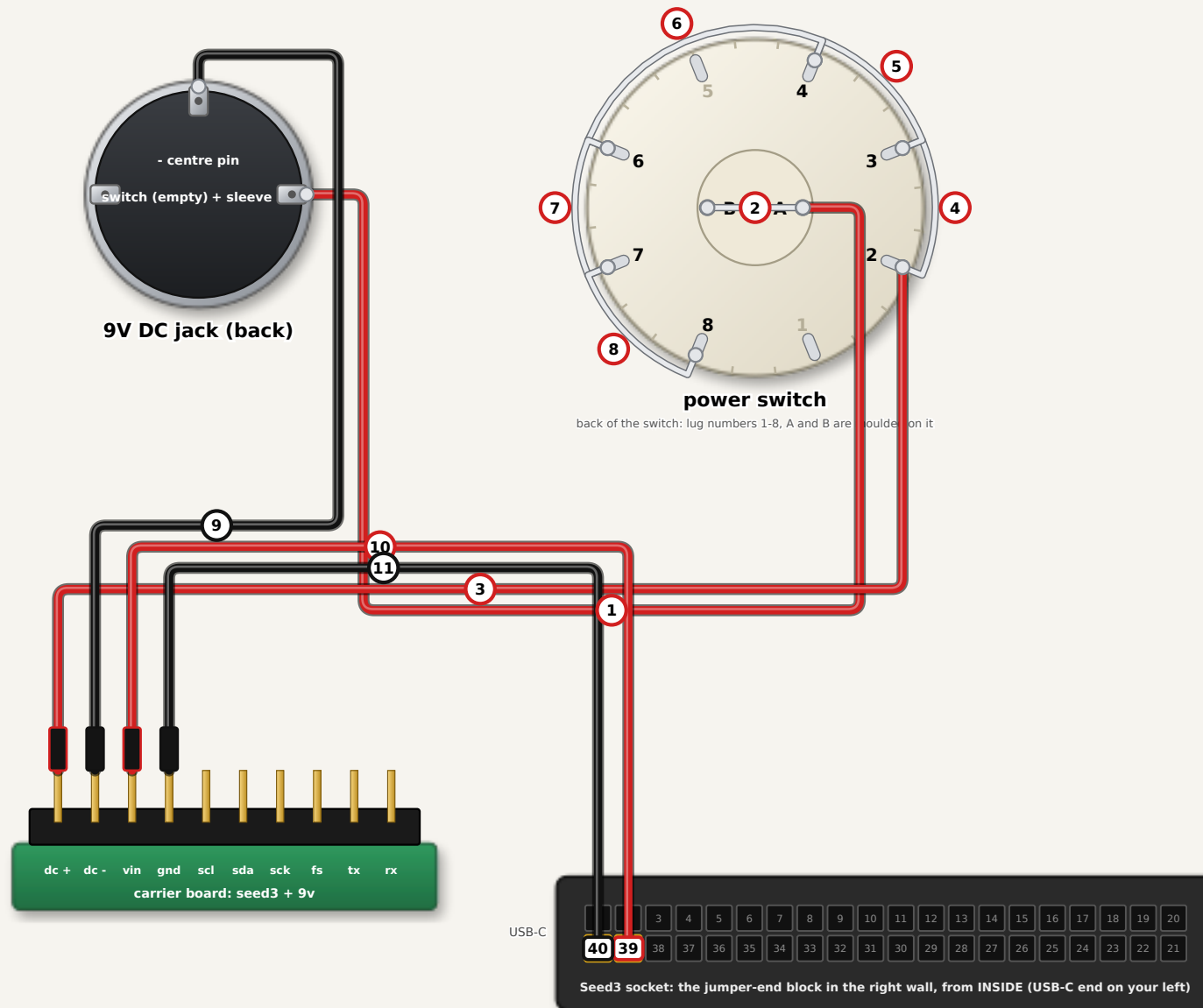


PG-1 wiring 1/10: Power: 9V jack -> power switch -> carrier board -> Seed3

+ (red) goes into BOTH commons, A and B. Lugs 2 3 4 and 6 7 8 are all ON: link them with bare offcuts (the short silver links, bent round the OUTSIDE so they never touch lug 1 or 5) and run ONE red wire from them to 'dc +'. Lugs 1 and 5 = OFF: leave them empty.

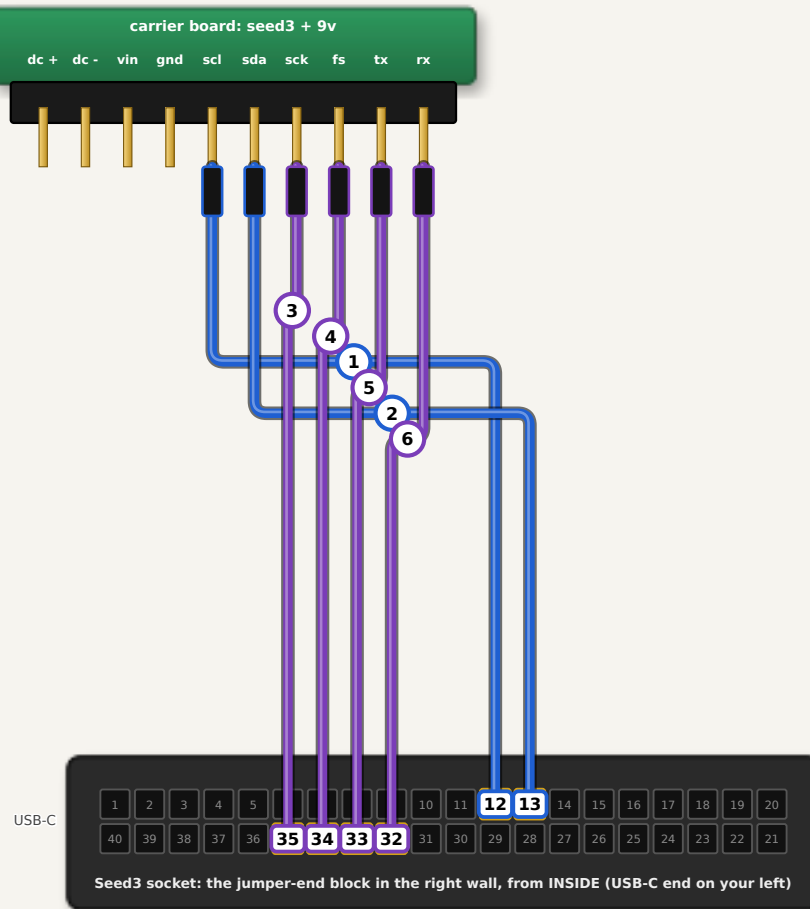


Steps (the number is on the wire)

- 1 9V jack: the SLEEVE lug, on the side: + -> power switch: common A (the inner lug by the moulded 'A') 9 V + into the switch (solder)
- 2 power switch: common A (the inner lug by the moulded 'A') -> power switch: common B (the inner lug by the moulded 'B') join both commons (short bare offcut)
- 3 power switch: lug 2 (pole A, position 2) -> carrier 'seed3 + 9v' header: 'dc +' (1st from the left) ON lugs -> dc + (join A2-A4 and B2-B4 with bare offcuts first)
- 4 power switch: lug 2 (pole A, position 2) -> power switch: lug 3 (pole A, position 3) join (bare offcut)
- 5 power switch: lug 3 (pole A, position 3) -> power switch: lug 4 (pole A, position 4) join (bare offcut)
- 6 power switch: lug 4 (pole A, position 4) -> power switch: lug 6 (pole B, position 2) join (bare offcut)
- 7 power switch: lug 6 (pole B, position 2) -> power switch: lug 7 (pole B, position 3) join (bare offcut)
- 8 power switch: lug 7 (pole B, position 3) -> power switch: lug 8 (pole B, position 4) join (bare offcut)
- 9 9V jack: the CENTRE-PIN lug, the middle one: - (centre negative) -> carrier 'seed3 + 9v' header: 'dc -' (2nd from the left) 9 V - straight to the board (female end on the header)
- 10 carrier 'seed3 + 9v' header: 'vin' (3rd from the left) -> Seed3 socket place 39 (bottom row) jumper: Seed3 power (9 V, protected)
- 11 carrier 'seed3 + 9v' header: 'gnd' (4th from the left) -> Seed3 socket place 40 (bottom row) jumper: Seed3 ground (the ground chain is spliced onto this wire)

PG-1 wiring 2/10: Carrier board -> Seed3: the 6 signal jumpers

Plug these onto the board's header BEFORE the board goes in the box (the names are printed on the board). Their other ends go into the Seed3 socket places shown.

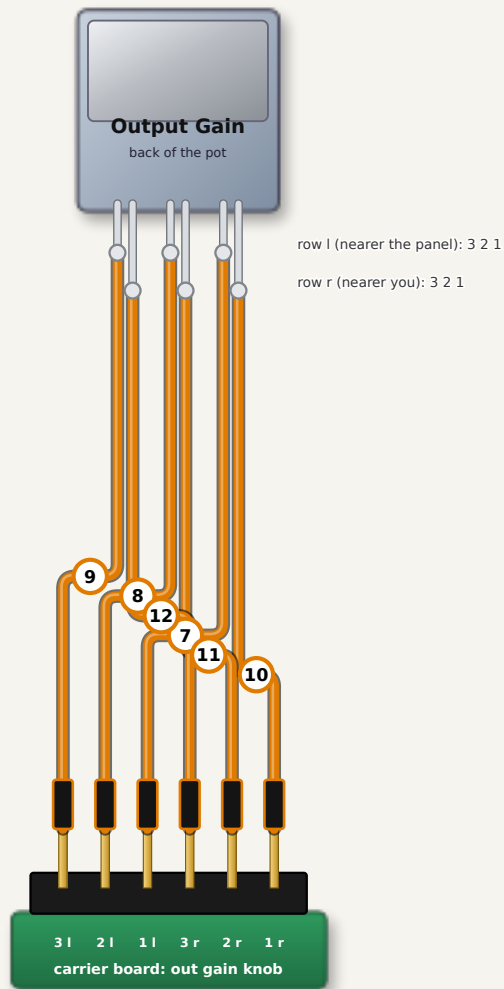
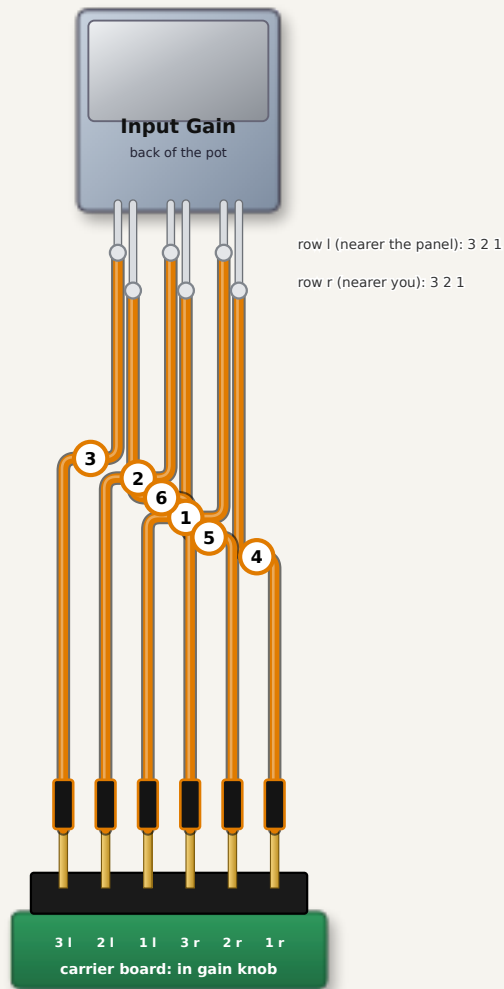


Steps (the number is on the wire)

- 1 carrier 'seed3 + 9v' header: 'scl' (5th from the left)
-> Seed3 socket place 12 (top row)
jumper: I2C clock
- 2 carrier 'seed3 + 9v' header: 'sda' (6th from the left)
-> Seed3 socket place 13 (top row)
jumper: I2C data
- 3 carrier 'seed3 + 9v' header: 'sck' (7th from the left)
-> Seed3 socket place 35 (bottom row)
jumper: audio bit clock
- 4 carrier 'seed3 + 9v' header: 'fs' (8th from the left)
-> Seed3 socket place 34 (bottom row)
jumper: audio frame clock
- 5 carrier 'seed3 + 9v' header: 'tx' (9th from the left)
-> Seed3 socket place 33 (bottom row)
jumper: audio Seed3 -> board
- 6 carrier 'seed3 + 9v' header: 'rx' (last on the right)
-> Seed3 socket place 32 (bottom row)
jumper: audio board -> Seed3

PG-1 wiring 3/10: The gain pots -> their headers on the carrier board

Short wires soldered on each pot, a female jumper end on the header. The pot's row nearer the PANEL is 'l', the row nearer YOU is 'r'. The header names are printed on the board.

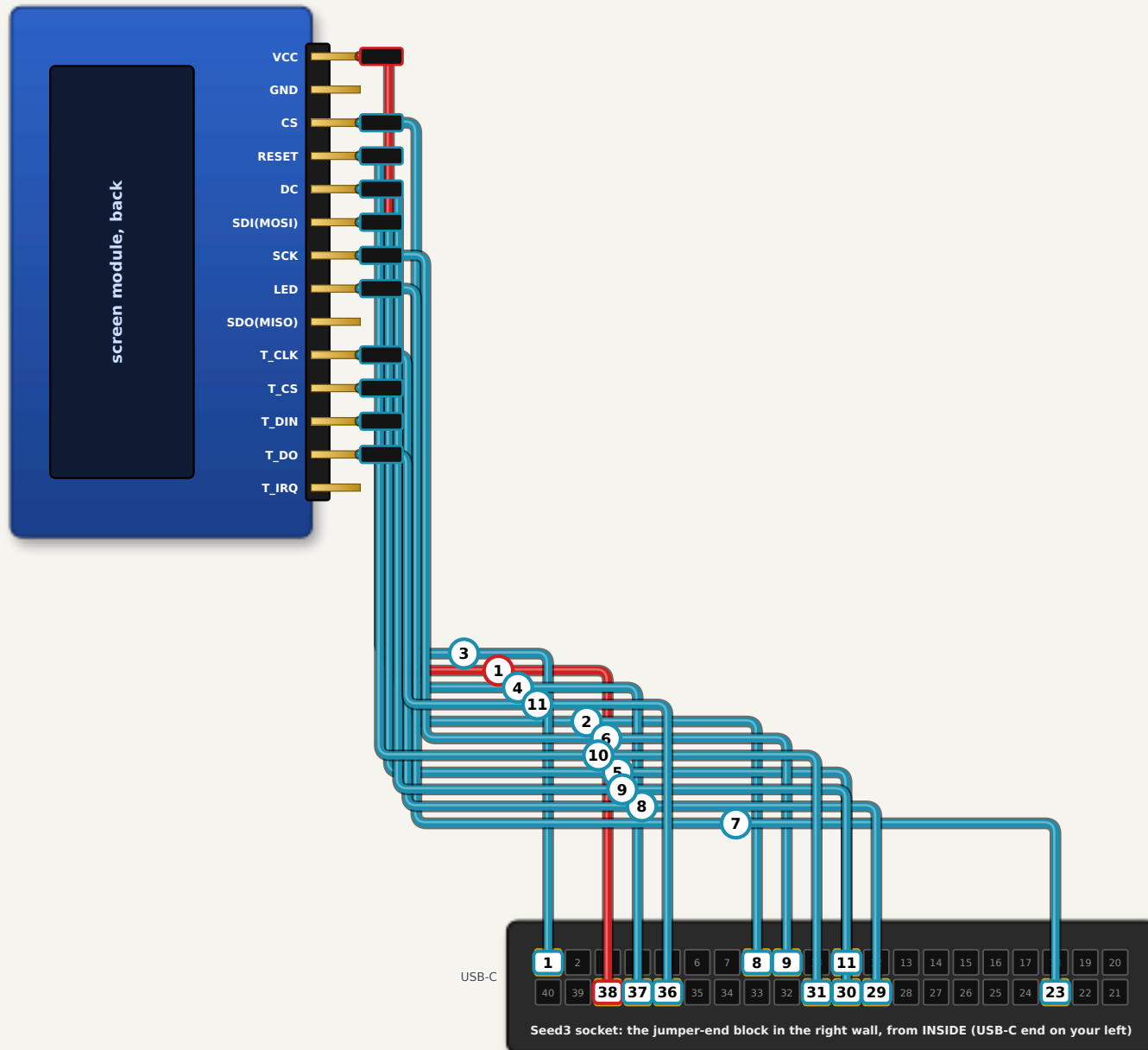


Steps (the number is on the wire)

- 1 Input Gain: pin 1 l (RIGHT pin of the row nearer the PANEL) -> carrier 'in gain knob' header: '1 l' (3rd from the left)
- 2 Input Gain: pin 2 l (MIDDLE pin of the row nearer the PANEL) -> carrier 'in gain knob' header: '2 l' (2nd from the left)
- 3 Input Gain: pin 3 l (LEFT pin of the row nearer the PANEL) -> carrier 'in gain knob' header: '3 l' (1st from the left)
- 4 Input Gain: pin 1 r (RIGHT pin of the row nearer YOU) -> carrier 'in gain knob' header: '1 r' (last on the right)
- 5 Input Gain: pin 2 r (MIDDLE pin of the row nearer YOU) -> carrier 'in gain knob' header: '2 r' (5th from the left)
- 6 Input Gain: pin 3 r (LEFT pin of the row nearer YOU) -> carrier 'in gain knob' header: '3 r' (4th from the left)
- 7 Output Gain: pin 1 l (RIGHT pin of the row nearer the PANEL) -> carrier 'out gain knob' header: '1 l' (3rd from the left)
- 8 Output Gain: pin 2 l (MIDDLE pin of the row nearer the PANEL) -> carrier 'out gain knob' header: '2 l' (2nd from the left)
- 9 Output Gain: pin 3 l (LEFT pin of the row nearer the PANEL) -> carrier 'out gain knob' header: '3 l' (1st from the left)
- 10 Output Gain: pin 1 r (RIGHT pin of the row nearer YOU) -> carrier 'out gain knob' header: '1 r' (last on the right)
- 11 Output Gain: pin 2 r (MIDDLE pin of the row nearer YOU) -> carrier 'out gain knob' header: '2 r' (5th from the left)
- 12 Output Gain: pin 3 r (LEFT pin of the row nearer YOU) -> carrier 'out gain knob' header: '3 r' (4th from the left)

PG-1 wiring 4/10: Screen -> Seed3

Read the names printed beside the screen's pins. 'SDO(MISO)' and 'T_IRQ' stay empty; 'GND' is on the ground-chain page.

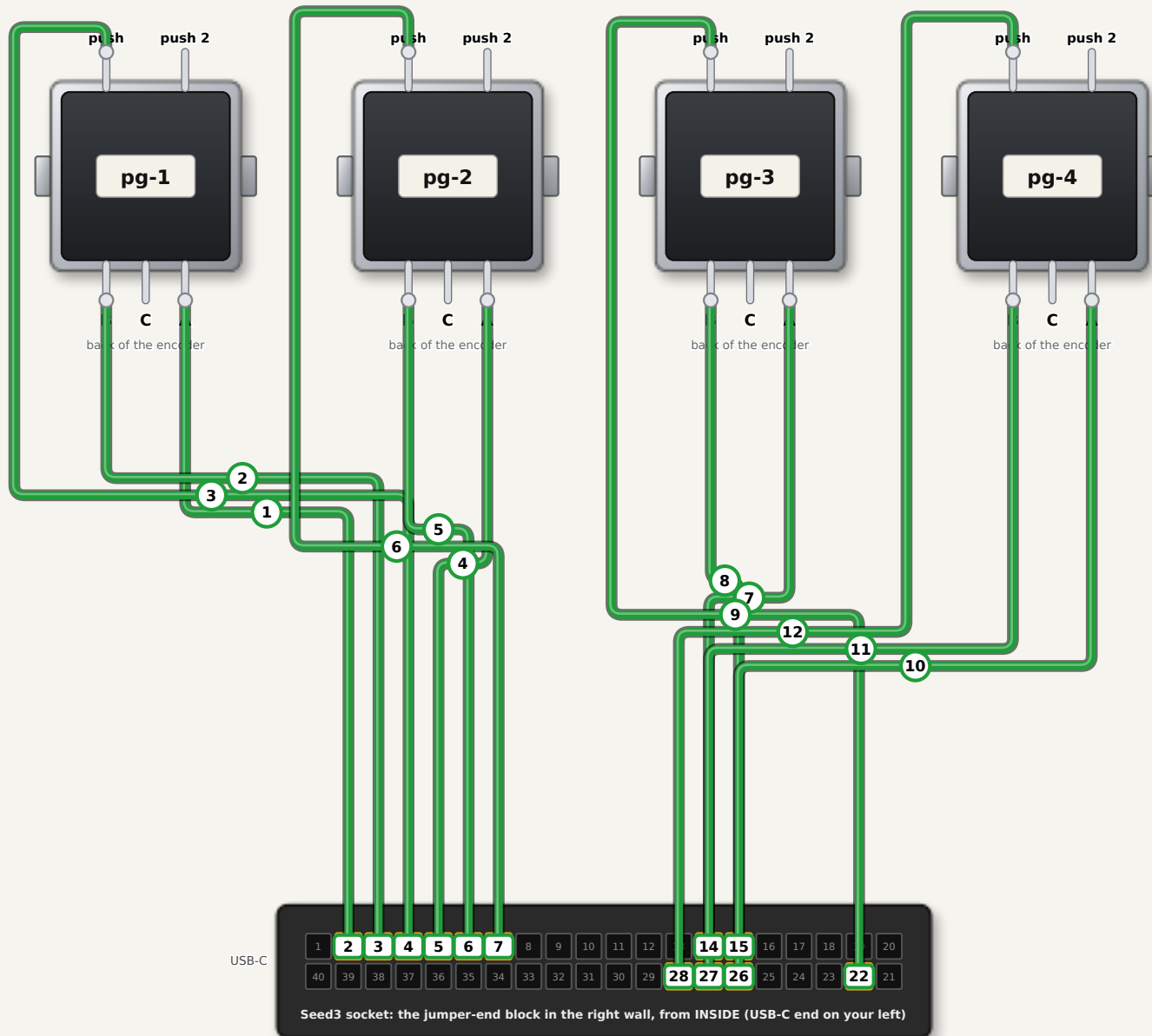


Steps (the number is on the wire)

- 1 screen: the pin printed 'VCC' (pin 1) -> Seed3 socket place 38 (bottom row)
jumper: 3.3 V
- 2 screen: the pin printed 'CS' (pin 3) -> Seed3 socket place 8 (top row)
- 3 screen: the pin printed 'RESET' (pin 4) -> Seed3 socket place 1 (top row)
- 4 screen: the pin printed 'DC' (pin 5) -> Seed3 socket place 37 (bottom row)
- 5 screen: the pin printed 'SDI(MOSI)' (pin 6) -> Seed3 socket place 11 (top row)
- 6 screen: the pin printed 'SCK' (pin 7) -> Seed3 socket place 9 (top row)
- 7 screen: the pin printed 'LED' (pin 8) -> Seed3 socket place 23 (bottom row)
jumper: backlight
- 8 screen: the pin printed 'T_CLK' (pin 10) -> Seed3 socket place 29 (bottom row)
jumper: touch
- 9 screen: the pin printed 'T_CS' (pin 11) -> Seed3 socket place 30 (bottom row)
jumper: touch
- 10 screen: the pin printed 'T_DIN' (pin 12) -> Seed3 socket place 31 (bottom row)
jumper: touch
- 11 screen: the pin printed 'T_DO' (pin 13) -> Seed3 socket place 36 (bottom row)
jumper: touch

PG-1 wiring 5/10: The 4 encoders -> Seed3

On the back of each encoder: a row of 3 pins and a row of 2. C and 'push 2' are on the ground-chain page. If a knob turns the wrong way, flip 'knobs' in the config tab: no rewiring.

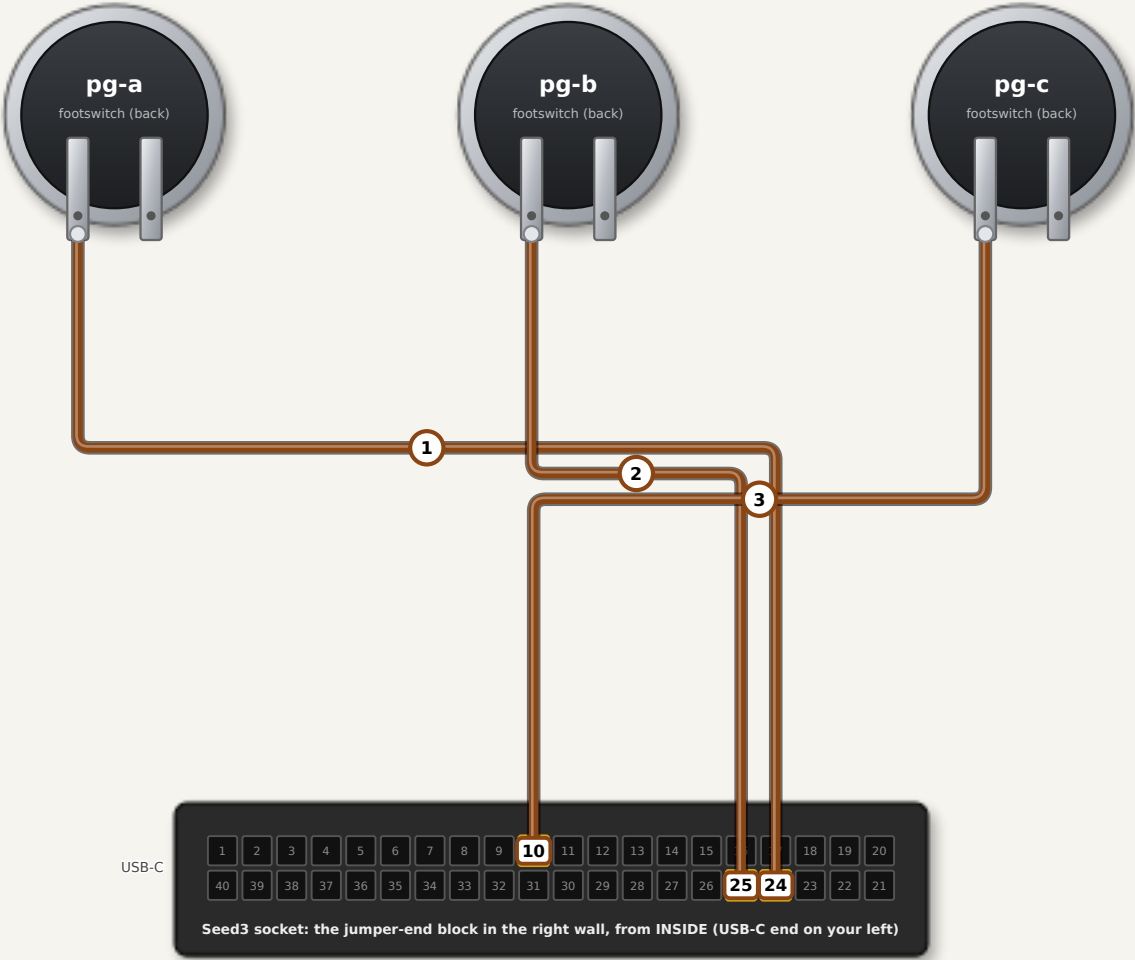


Steps (the number is on the wire)

- 1 pg-1 encoder: RIGHT pin of the row of 3 (A) -> Seed3 socket place 2 (top row)
- 2 pg-1 encoder: LEFT pin of the row of 3 (B) -> Seed3 socket place 3 (top row)
- 3 pg-1 encoder: LEFT pin of the 2 (push) -> Seed3 socket place 4 (top row)
- 4 pg-2 encoder: RIGHT pin of the row of 3 (A) -> Seed3 socket place 5 (top row)
- 5 pg-2 encoder: LEFT pin of the row of 3 (B) -> Seed3 socket place 6 (top row)
- 6 pg-2 encoder: LEFT pin of the 2 (push) -> Seed3 socket place 7 (top row)
- 7 pg-3 encoder: RIGHT pin of the row of 3 (A) -> Seed3 socket place 14 (top row)
- 8 pg-3 encoder: LEFT pin of the row of 3 (B) -> Seed3 socket place 15 (top row)
- 9 pg-3 encoder: LEFT pin of the 2 (push) -> Seed3 socket place 22 (bottom row)
- 10 pg-4 encoder: RIGHT pin of the row of 3 (A) -> Seed3 socket place 26 (bottom row)
- 11 pg-4 encoder: LEFT pin of the row of 3 (B) -> Seed3 socket place 27 (bottom row)
- 12 pg-4 encoder: LEFT pin of the 2 (push) -> Seed3 socket place 28 (bottom row)

PG-1 wiring 6/10: The 3 footswitches -> Seed3

Either lug of each footswitch goes to its Seed3 place; the other lug is on the ground-chain page.

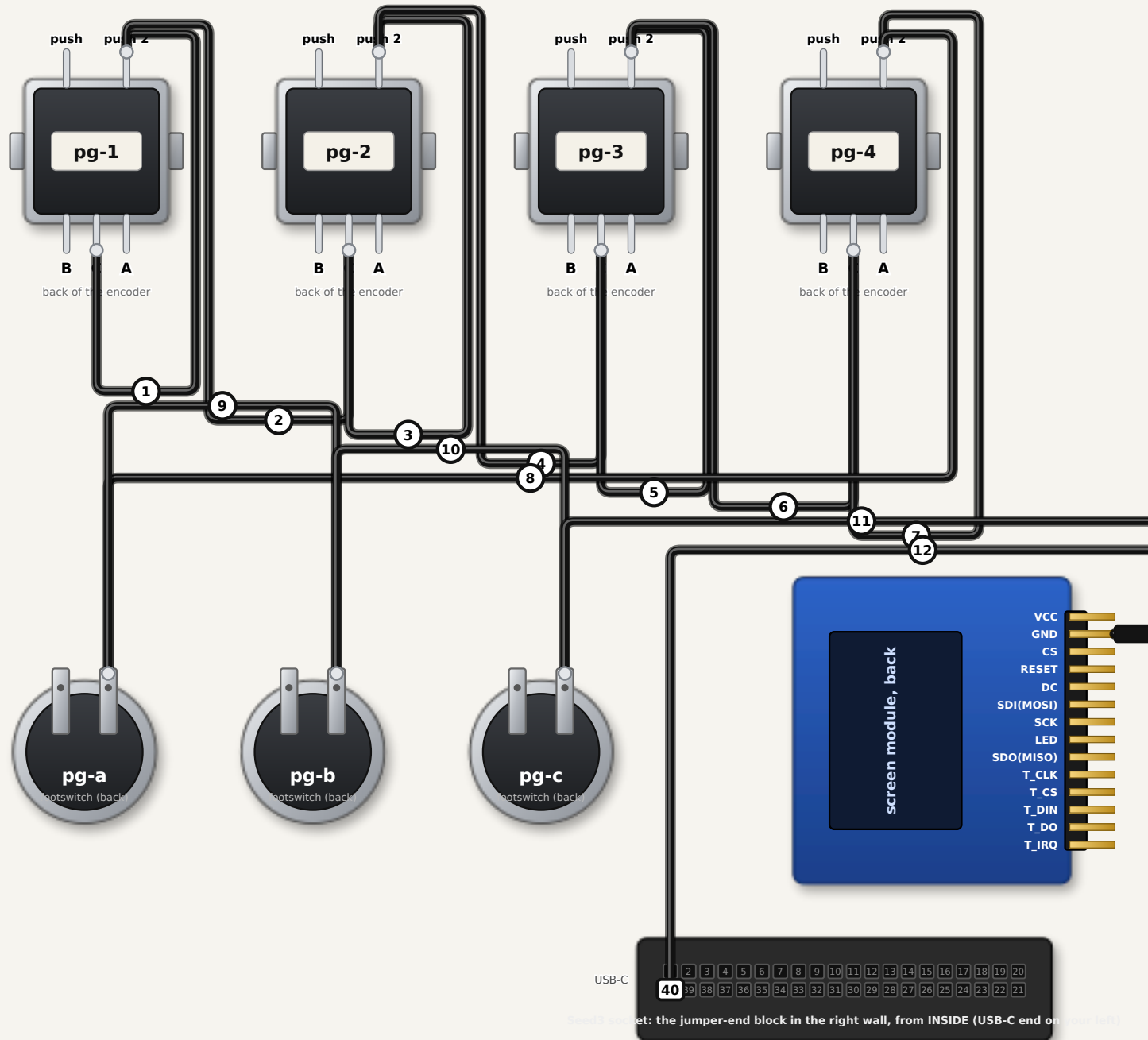


Steps (the number is on the wire)

- 1 pg-a footswitch: either lug -> Seed3 socket place 24 (bottom row)
- 2 pg-b footswitch: either lug -> Seed3 socket place 25 (bottom row)
- 3 pg-c footswitch: either lug -> Seed3 socket place 10 (top row)

PG-1 wiring 7/10: The ground chain (black wire, part to part)

One black wire from each lug to the next, in step order. The last one (from the screen's GND) is **SPLICED** onto the jumper that runs from the board's 'gnd' to Seed3 place 40, about 3 cm from the socket: strip 5 mm, wrap, solder, heat shrink (a socket place only holds one jumper end).

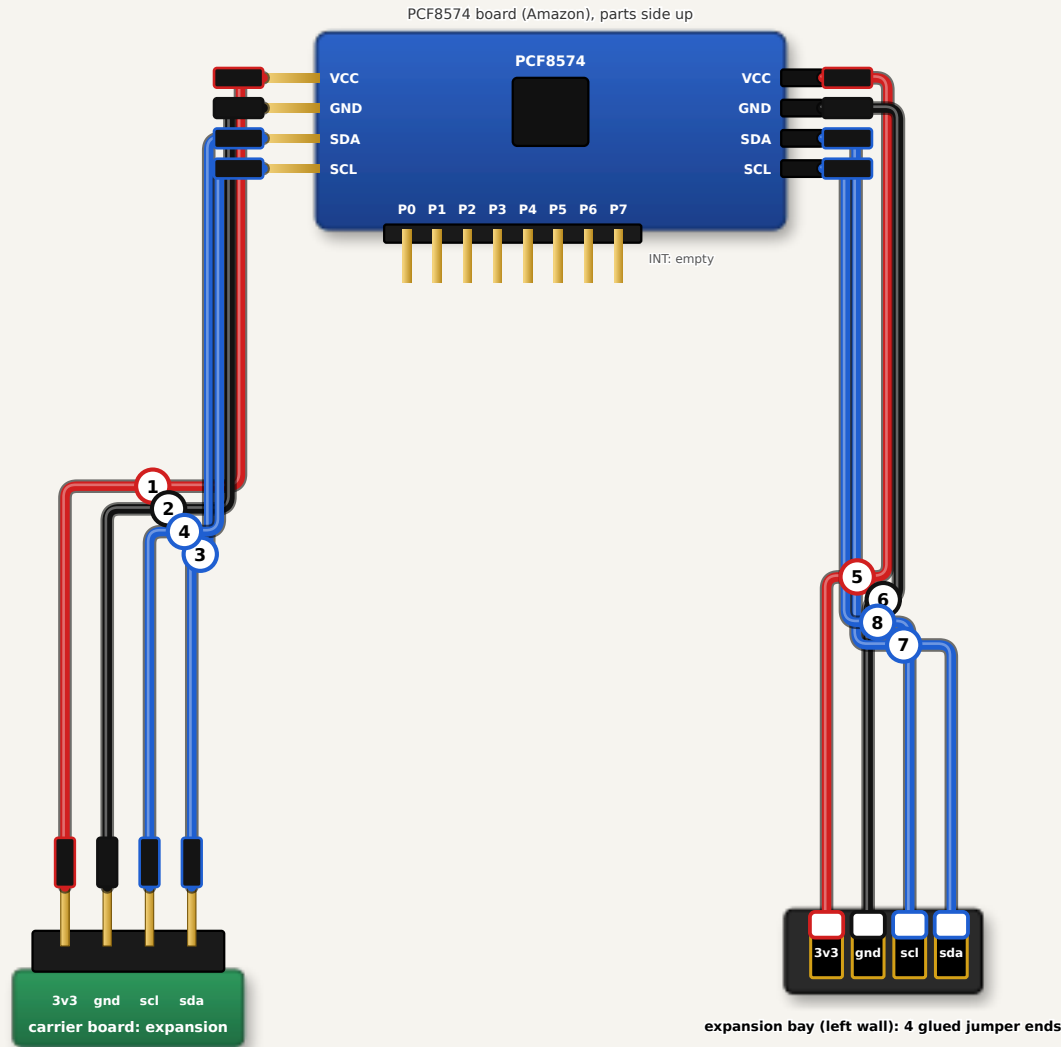


Steps (the number is on the wire)

- 1 pg-1 encoder: MIDDLE pin of the row of 3 (C) -> pg-1 encoder: RIGHT pin of the 2 (push 2)
- 2 pg-1 encoder: RIGHT pin of the 2 (push 2) -> pg-2 encoder: MIDDLE pin of the row of 3 (C)
- 3 pg-2 encoder: MIDDLE pin of the row of 3 (C) -> pg-2 encoder: RIGHT pin of the 2 (push 2)
- 4 pg-2 encoder: RIGHT pin of the 2 (push 2) -> pg-3 encoder: MIDDLE pin of the row of 3 (C)
- 5 pg-3 encoder: MIDDLE pin of the row of 3 (C) -> pg-3 encoder: RIGHT pin of the 2 (push 2)
- 6 pg-3 encoder: RIGHT pin of the 2 (push 2) -> pg-4 encoder: MIDDLE pin of the row of 3 (C)
- 7 pg-4 encoder: MIDDLE pin of the row of 3 (C) -> pg-4 encoder: RIGHT pin of the 2 (push 2)
- 8 pg-4 encoder: RIGHT pin of the 2 (push 2) -> pg-a footswitch: the other lug
- 9 pg-a footswitch: the other lug -> pg-b footswitch: the other lug
- 10 pg-b footswitch: the other lug -> pg-c footswitch: the other lug
- 11 pg-c footswitch: the other lug -> screen: the pin printed 'GND' (pin 2)
ground chain (screen GND: cut a jumper, solder its cut end in the chain)
- 12 screen: the pin printed 'GND' (pin 2) -> Seed3 socket place 40 (bottom row)
chain's end: SPLICED onto the j10.gnd -> seed.40 jumper, 3 cm from the socket (strip 5 mm, wrap, solder, heat shrink). A socket place holds one jumper only

PG-1 wiring 8/10: PCF8574 board -> carrier board and the expansion bay

The 4 input pins go to the board's 'expansion' header; the 4 pass-through pins go to the bay in the left wall. The PCF8574 board sits on the inside of the lid (velcro).

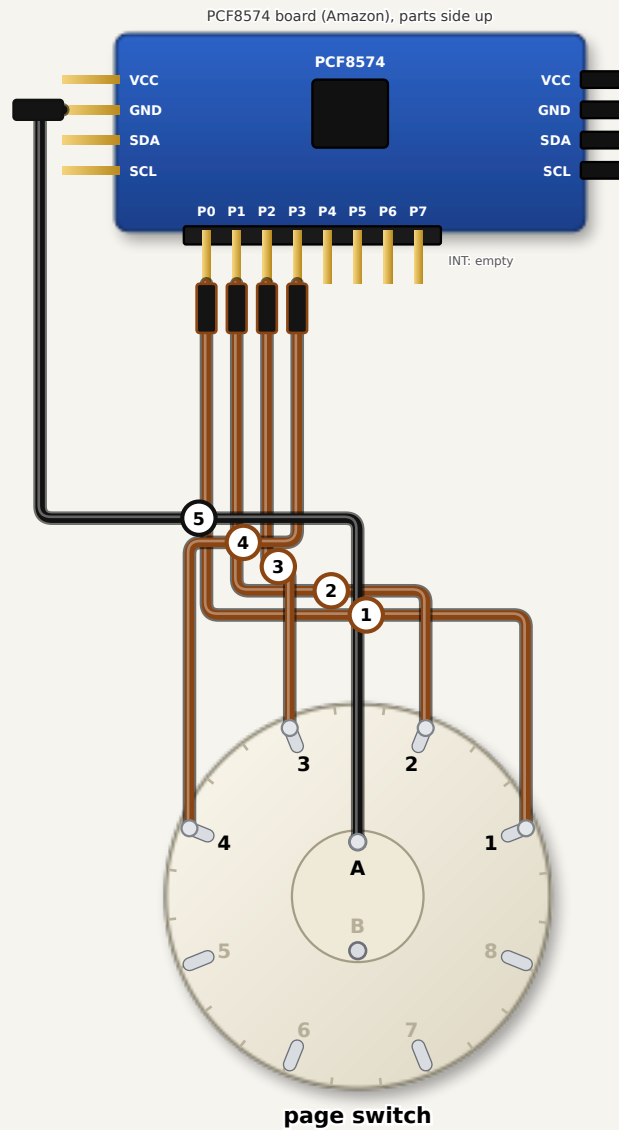


Steps (the number is on the wire)

- 1 PCF8574: 'VCC' on the INPUT pins (left end) -> carrier 'expansion' header: '3v3' (1st from the left) jumper (3.3 V)
- 2 PCF8574: 'GND' on the INPUT pins (left end) -> carrier 'expansion' header: 'gnd' (2nd from the left)
- 3 PCF8574: 'SDA' on the INPUT pins (left end) -> carrier 'expansion' header: 'sda' (last on the right)
- 4 PCF8574: 'SCL' on the INPUT pins (left end) -> carrier 'expansion' header: 'scl' (3rd from the left)
- 5 PCF8574: 'VCC' on the PASS-THROUGH pins (right end) -> expansion bay: the '3v3' jumper end jumper, its far end glued in the bay
- 6 PCF8574: 'GND' on the PASS-THROUGH pins (right end) -> expansion bay: the 'gnd' jumper end jumper, its far end glued in the bay
- 7 PCF8574: 'SDA' on the PASS-THROUGH pins (right end) -> expansion bay: the 'sda' jumper end jumper, its far end glued in the bay
- 8 PCF8574: 'SCL' on the PASS-THROUGH pins (right end) -> expansion bay: the 'scl' jumper end jumper, its far end glued in the bay

PG-1 wiring 9/10: Page switch -> PCF8574 board

Pole A only: lugs 1 2 3 4 and common A. Pole B (5 6 7 8 and B) stays empty. Common A goes to GND; that pin already has the carrier's jumper on it, so solder this one on the PCF board's **UNDERSIDE**, on GND's joint.



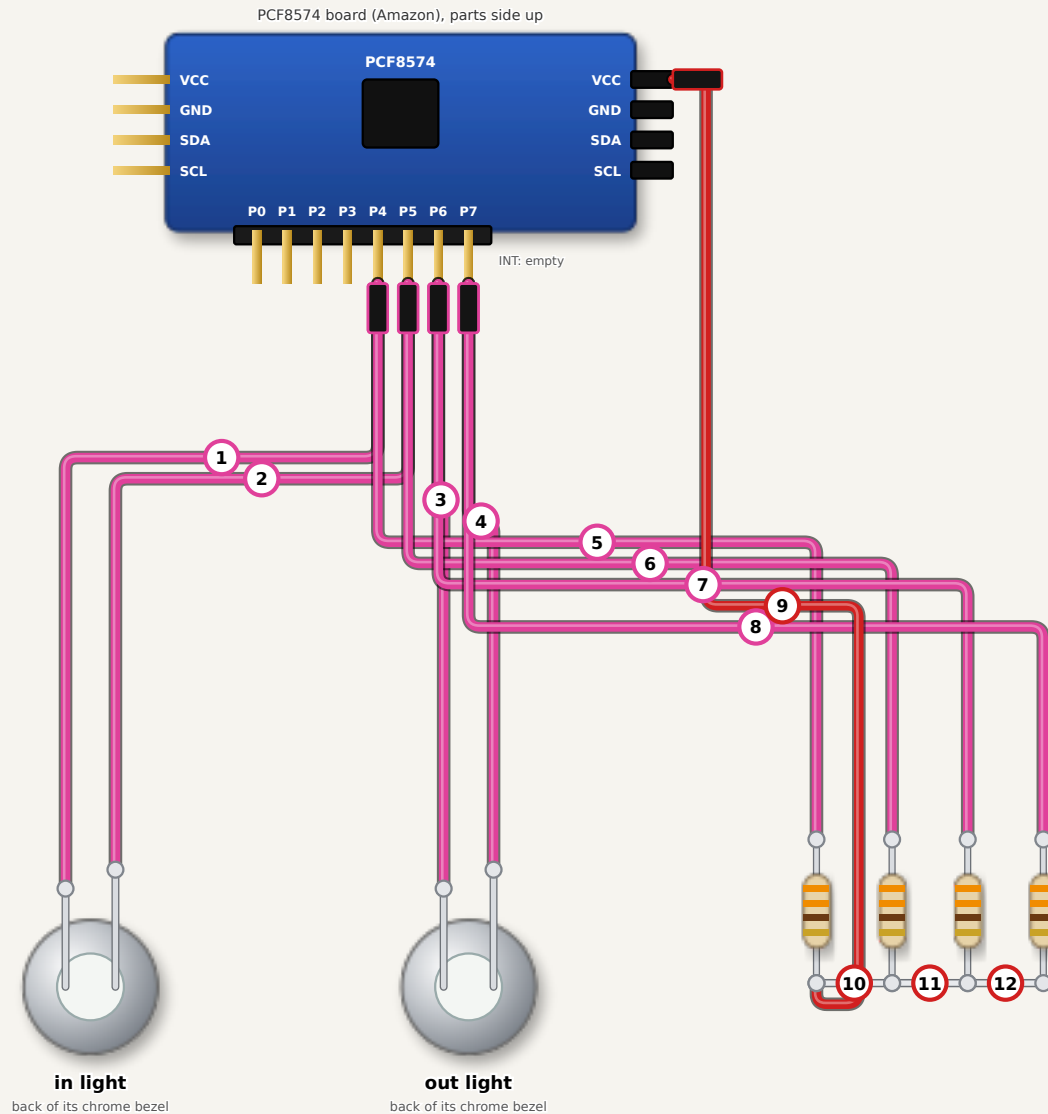
back of the switch: lug numbers 1-8, A and B are moulded on it; drawn turned (it fits any way round)

Steps (the number is on the wire)

- 1 page switch: lug 1 (pole A, position 1) -> PCF8574: 'P0' on the row of 8 position 1 (find with beep mode)
- 2 page switch: lug 2 (pole A, position 2) -> PCF8574: 'P1' on the row of 8 position 2
- 3 page switch: lug 3 (pole A, position 3) -> PCF8574: 'P2' on the row of 8 position 3
- 4 page switch: lug 4 (pole A, position 4) -> PCF8574: 'P3' on the row of 8 position 4
- 5 page switch: common A (the inner lug by the moulded 'A') -> PCF8574: 'GND' on the INPUT pins (left end)
pole A common: solder it on the **UNDERSIDE** of the PCF board, on the GND pin's solder joint

PG-1 wiring 10/10: Level lights + 330 ohm resistors -> PCF8574 board

Each light's 2 legs go to 2 P pins, either way round (red with signal? swap that light's 2 wires). P4-P7 each also get a 330 ohm to VCC, soldered on the PCF board's UNDERSIDE; the 4 resistors' bottom legs are joined and go to a VCC pin.



Steps (the number is on the wire)

- 1 in light: the LEFT leg (either way round) -> PCF8574: 'P4' on the row of 8
in light (swap its 2 legs, or set kSwapLightColours, if the colours are backwards)
- 2 in light: the RIGHT leg (either way round) -> PCF8574: 'P5' on the row of 8
in light
- 3 out light: the LEFT leg (either way round) -> PCF8574: 'P6' on the row of 8
out light
- 4 out light: the RIGHT leg (either way round) -> PCF8574: 'P7' on the row of 8
out light
- 5 330 ohm resistor 1: its TOP leg -> PCF8574: 'P4' on the row of 8
330 ohm pull-up: solder its leg on the PCF board's underside, on P4's joint
- 6 330 ohm resistor 2: its TOP leg -> PCF8574: 'P5' on the row of 8
330 ohm pull-up
- 7 330 ohm resistor 3: its TOP leg -> PCF8574: 'P6' on the row of 8
330 ohm pull-up
- 8 330 ohm resistor 4: its TOP leg -> PCF8574: 'P7' on the row of 8
330 ohm pull-up
- 9 330 ohm resistor 1: its BOTTOM leg -> PCF8574: 'VCC' on the PASS-THROUGH pins (right end)
the 4 resistors' other ends joined -> underside of a VCC pin's joint
- 10 330 ohm resistor 1: its BOTTOM leg -> 330 ohm resistor 2: its BOTTOM leg
join
- 11 330 ohm resistor 2: its BOTTOM leg -> 330 ohm resistor 3: its BOTTOM leg
join
- 12 330 ohm resistor 3: its BOTTOM leg -> 330 ohm resistor 4: its BOTTOM leg
join